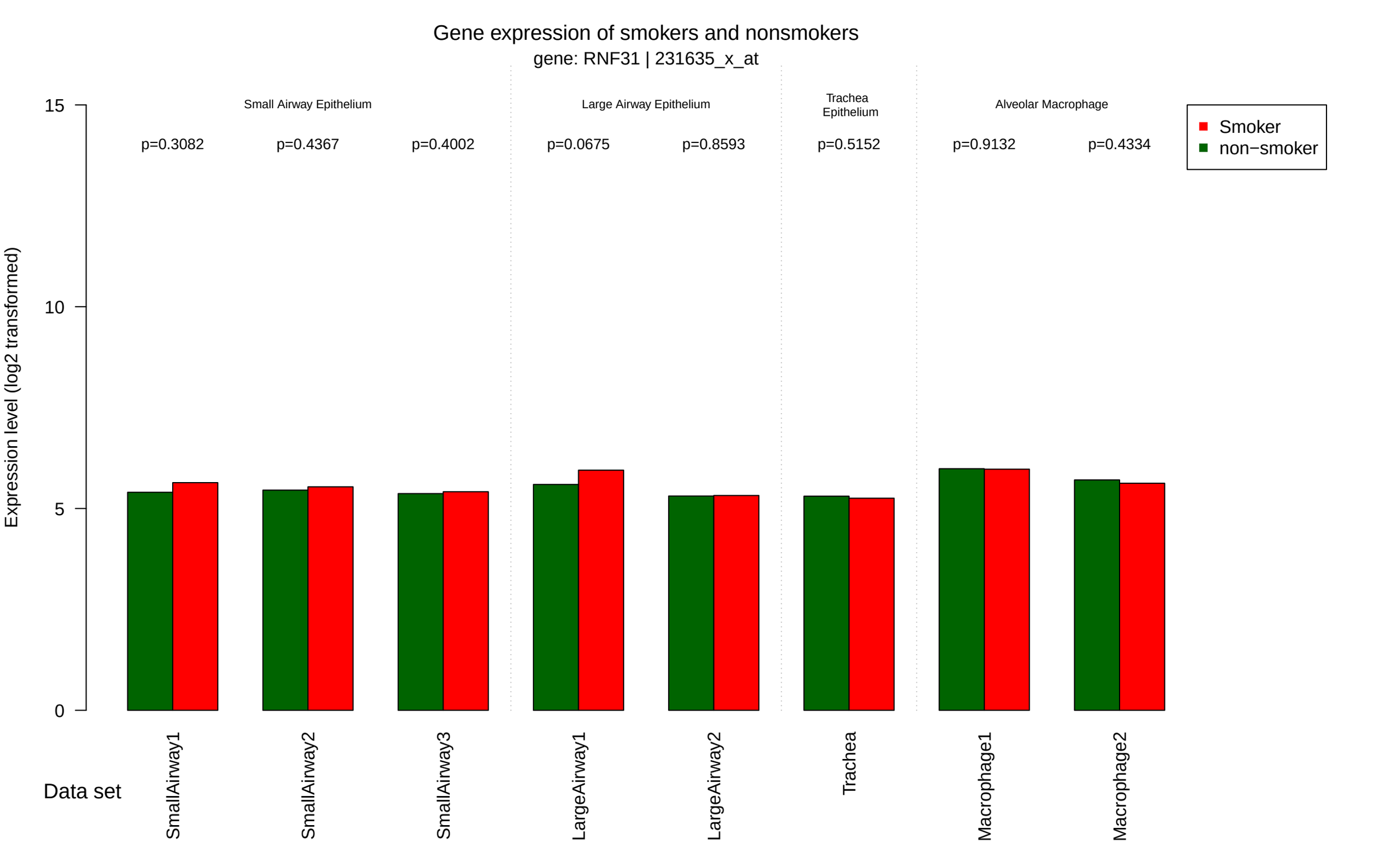
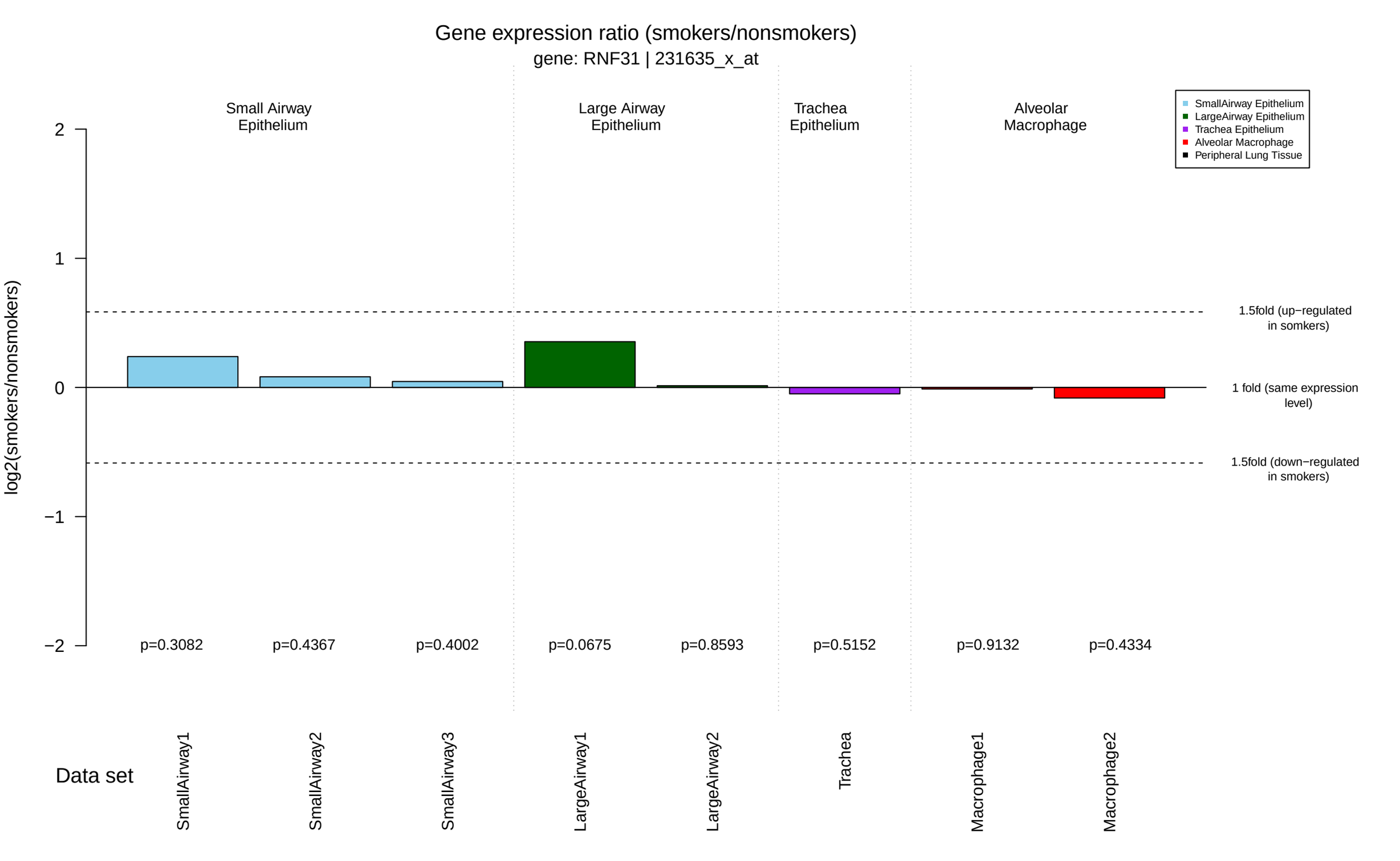
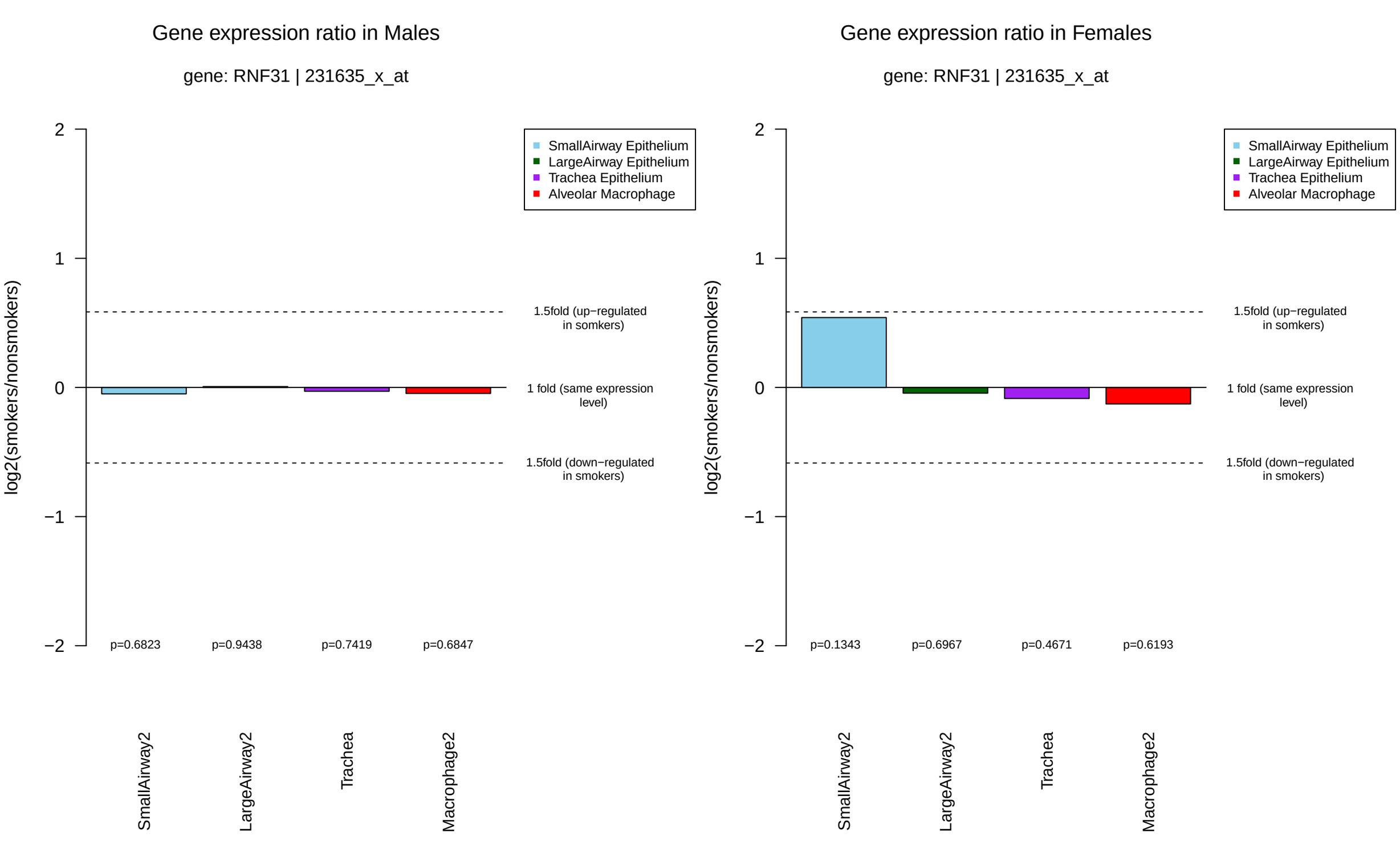
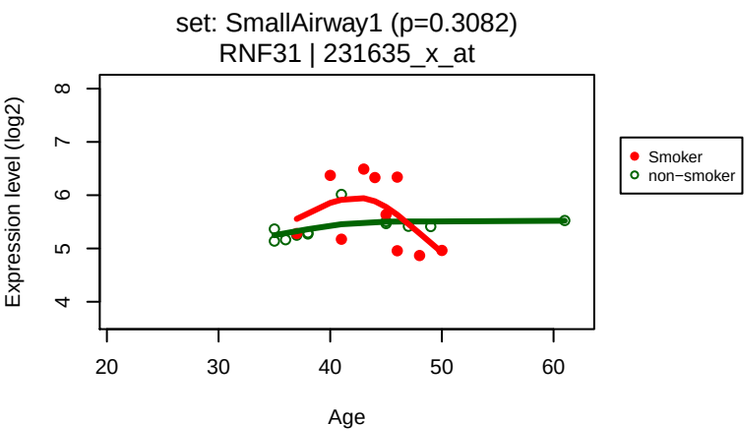
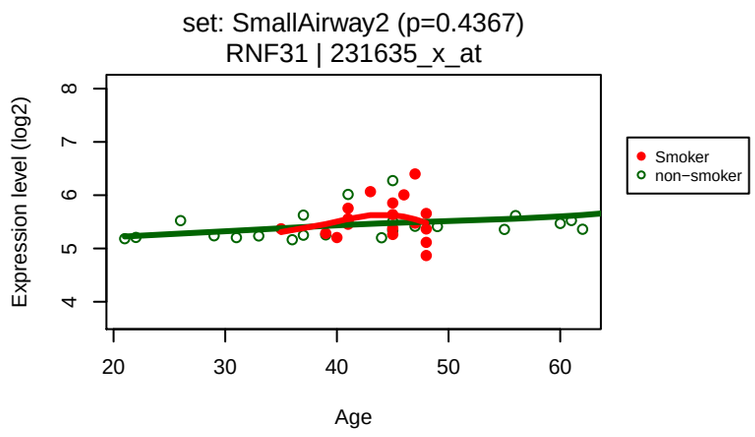
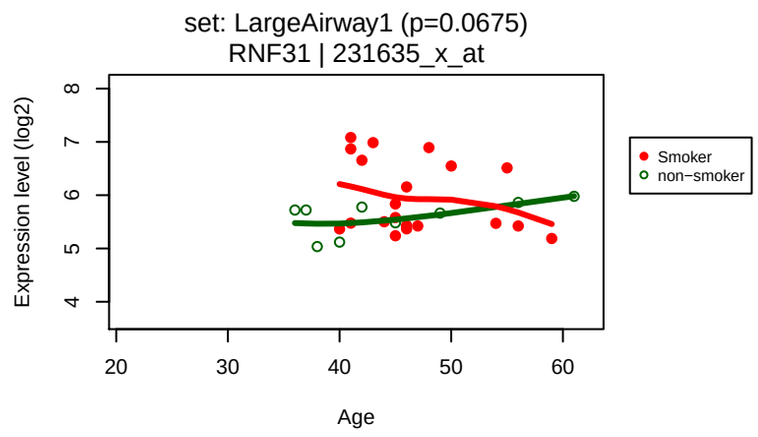

Gene expression of smokers and nonsmokers
gene: RNF31 | 231635_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3082
p=0.4367
p=0.4002
p=0.0675
p=0.8593
p=0.5152
p=0.9132
p=0.4334
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

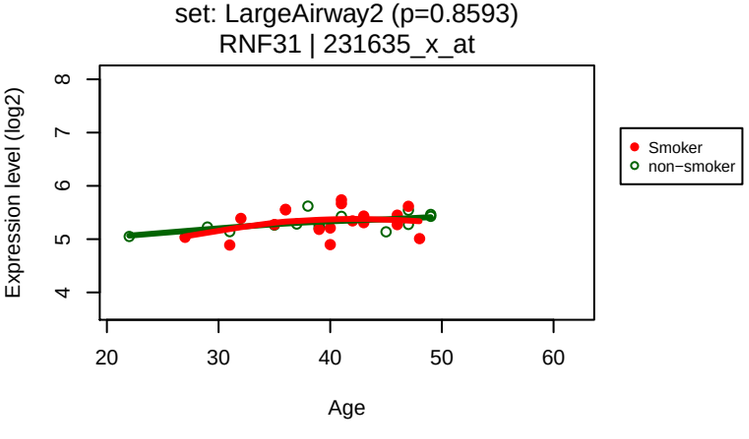

Gene expression ratio (smokers/nonsmokers)
gene: RNF31 | 231635_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3082
p=0.4367
p=0.4002
p=0.0675
p=0.8593
p=0.5152
p=0.9132
p=0.4334
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

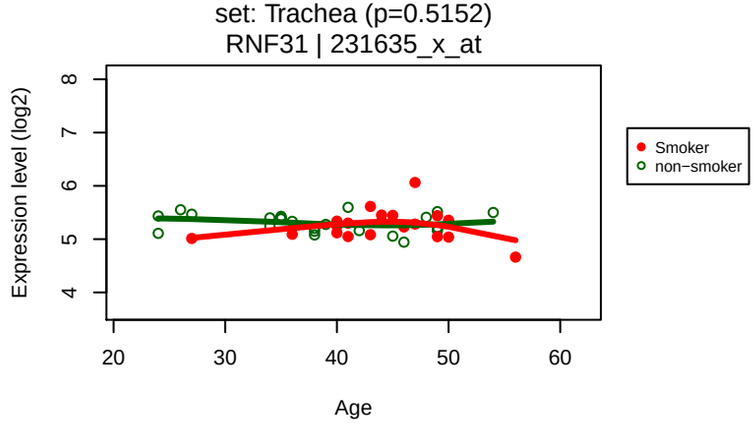

Gene expression ratio in Males
gene: RNF31 | 231635_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6823
p=0.9438
p=0.7419
p=0.6847
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RNF31 | 231635_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1343
p=0.6967
p=0.4671
p=0.6193
SmallAirway2
LargeAirway2
Trachea
Macrophage2

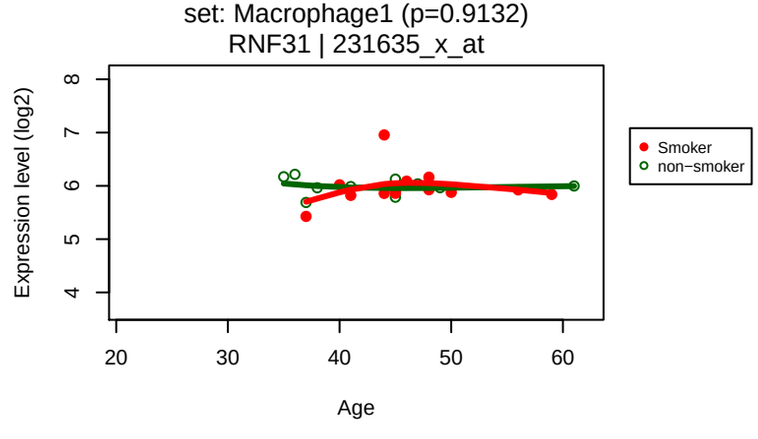

set: SmallAirway1 (p=0.3082)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker

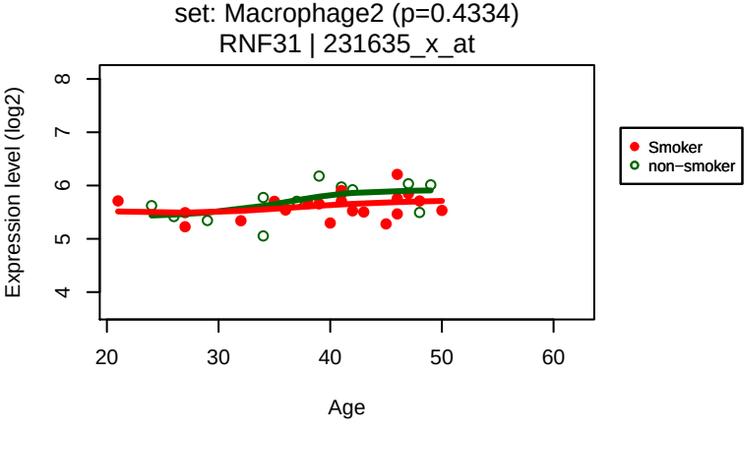

set: SmallAirway2 (p=0.4367)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker

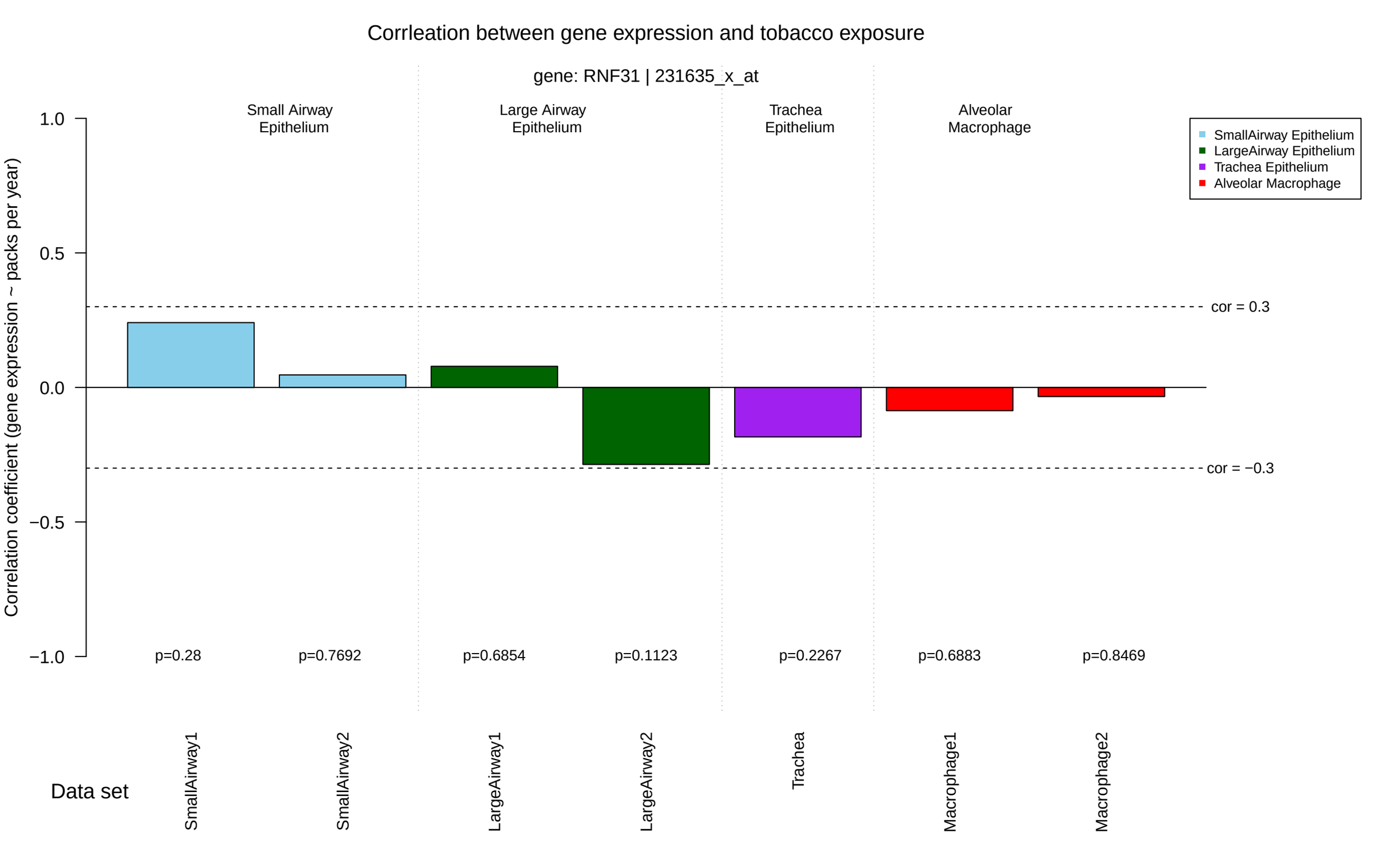

set: LargeAirway1 (p=0.0675)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8593)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5152)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9132)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4334)
RNF31 | 231635_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RNF31 | 231635_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.28
p=0.7692
p=0.6854
p=0.1123
p=0.2267
p=0.6883
p=0.8469
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2